

Part A: Define Your Multivariate Normal Distribution

Mean Vector:

$$\boldsymbol{\mu} = \begin{bmatrix} 20 \\ 15 \\ 10 \end{bmatrix}$$

Standard Deviations:

$$\sigma_1 = 5, \quad \sigma_2 = 3, \quad \sigma_3 = 4.$$

Covariance Matrix:

$$\boldsymbol{\Sigma} = \begin{bmatrix} 25 & 3 & 15 \\ 3 & 9 & 7.8 \\ 15 & 7.8 & 16 \end{bmatrix}$$

Correlation Matrix:

$$R = \begin{bmatrix} 1.00 & 0.20 & 0.75 \\ 0.20 & 1.00 & 0.65 \\ 0.75 & 0.65 & 1.00 \end{bmatrix}$$

Variables:

- X_1 : Customer arrival rate (customers per hour)
- X_2 : Average service processing time (minutes)
- X_3 : Maximum queue length (number of customers)

Part B: Generate and Explore the Simulated Data

Sample size:

$$n = 1000$$

Random seed:

$$\text{seed} = 123$$

Table B.1: Theoretical and Simulated Means

Table 1: Theoretical and simulated means from Part B.		
Variable	Theoretical Mean	Simulated Mean
Customer Arrival Rate (X_1)	20	19.962999
Service Processing Time (X_2)	15	15.041682
Maximum Queue Length (X_3)	10	10.001995

Table 2: Theoretical and simulated variances from Part B.

Variable	Theoretical Variance	Simulated Variance
Customer Arrival Rate (X_1)	25	22.271714
Service Processing Time (X_2)	9	9.320508
Maximum Queue Length (X_3)	16	14.998812

Table B.2: Theoretical and Simulated Variances

Table B.3: Theoretical Covariance Matrix

$$\Sigma = \begin{bmatrix} 25 & 3 & 15 \\ 3 & 9 & 7.8 \\ 15 & 7.8 & 16 \end{bmatrix}$$

Table B.4: Simulated Covariance Matrix

$$\hat{\Sigma} = \begin{bmatrix} 22.271714 & 2.531737 & 13.430342 \\ 2.531737 & 9.320508 & 7.692923 \\ 13.430342 & 7.692923 & 14.998812 \end{bmatrix}$$

Table B.5: Theoretical Correlation Matrix

$$R = \begin{bmatrix} 1.00 & 0.20 & 0.75 \\ 0.20 & 1.00 & 0.65 \\ 0.75 & 0.65 & 1.00 \end{bmatrix}$$

Table B.6: Simulated Correlation Matrix

$$\hat{R} = \begin{bmatrix} 1.000 & 0.176 & 0.735 \\ 0.176 & 1.000 & 0.651 \\ 0.735 & 0.651 & 1.000 \end{bmatrix}$$

First five simulated observations:

Table 3: First five observations from the simulated sample in Part B.

Observation	X_1	X_2	X_3
0	26.654277	13.917300	13.139680
1	25.265506	17.482553	17.736759
2	30.003219	18.761054	20.653479
3	22.925063	18.057025	13.932207
4	12.142650	14.700125	4.791629

Part C: Investigate the Effect of Correlation

Variables examined:

$$X_1 = \text{Customer Arrival Rate}, \quad X_3 = \text{Maximum Queue Length}$$

Fixed parameters:

$$\mu_1 = 20, \quad \mu_3 = 10$$

$$\sigma_1 = 5, \quad \sigma_3 = 4$$

Sample size:

$$n = 1000$$

Table C.1: Correlation Settings

Table 4: Correlation settings investigated in Part C.

Setting	ρ
1	0.00
2	0.50
3	0.90

The covariance between X_1 and X_3 is calculated as

$$\text{Cov}(X_1, X_3) = \rho\sigma_1\sigma_3.$$

Therefore, the covariance values for each setting are:

$$\begin{aligned} \rho = 0.00 : \quad & \text{Cov}(X_1, X_3) = 0(5)(4) = 0, \\ \rho = 0.50 : \quad & \text{Cov}(X_1, X_3) = 0.50(5)(4) = 10, \\ \rho = 0.90 : \quad & \text{Cov}(X_1, X_3) = 0.90(5)(4) = 18. \end{aligned}$$

Table C.2: Covariance Matrices for Each Correlation Setting

$$\begin{aligned} \rho = 0.00 : \quad & \Sigma_0 = \begin{bmatrix} 25 & 0 \\ 0 & 16 \end{bmatrix} \\ \rho = 0.50 : \quad & \Sigma_{0.50} = \begin{bmatrix} 25 & 10 \\ 10 & 16 \end{bmatrix} \\ \rho = 0.90 : \quad & \Sigma_{0.90} = \begin{bmatrix} 25 & 18 \\ 18 & 16 \end{bmatrix} \end{aligned}$$

Part D: Investigate a Linear Combination

The selected linear combination is

$$Y = X_1 + 2X_2 - X_3.$$

The coefficient vector is

$$\mathbf{a} = \begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix}.$$

The theoretical mean and variance are

$$E(Y) = \mathbf{a}'\boldsymbol{\mu} = 40$$

and

$$\text{Var}(Y) = \mathbf{a}'\boldsymbol{\Sigma}\mathbf{a} = 27.8.$$

Thus,

$$Y \sim N(40, 27.8).$$

Table D.1: Theoretical and Simulated Results

Table 5: Theoretical and simulated results for the linear combination in Part D.

Measure	Theoretical	Simulated
Mean	40.000000	40.044367
Variance	27.800000	27.047131

The theoretical result is

$$Y = \mathbf{a}'X \sim N(\mathbf{a}'\boldsymbol{\mu}, \mathbf{a}'\boldsymbol{\Sigma}\mathbf{a})$$

and for this specific linear combination,

$$Y \sim N(40, 27.8).$$

Part E: Mahalanobis Distance and Chi-Square

The squared Mahalanobis distance is

$$D^2 = (\mathbf{x} - \boldsymbol{\mu})'\boldsymbol{\Sigma}^{-1}(\mathbf{x} - \boldsymbol{\mu}).$$

Since there are three variables,

$$D^2 \sim \chi_3^2.$$

Table 6: Mahalanobis distance results from Part E.

Measure	Value
Degrees of freedom	3
90% Critical Value	6.2514
Proportion Inside 90% Region	0.919
Percentage Inside 90% Region	91.9%

Table E.1: Mahalanobis Distance Results

The theoretical 90% critical value is

$$\chi_3^2(0.90) = 6.2514.$$

Approximately

$$91.9\%$$

of the observations had D^2 values below this critical value.
Observations satisfying

$$D^2 > 6.2514$$

are relatively far from the multivariate mean.

Custom Investigation: Option B – Subset of Variables

The subset consists of

$$X_1 = \text{Customer Arrival Rate}$$

and

$$X_2 = \text{Service Processing Time.}$$

The theoretical mean vector is

$$\boldsymbol{\mu}_{\text{sub}} = \begin{bmatrix} 20 \\ 15 \end{bmatrix}$$

and the theoretical covariance matrix is

$$\boldsymbol{\Sigma}_{\text{sub}} = \begin{bmatrix} 25 & 3 \\ 3 & 9 \end{bmatrix}.$$

Therefore,

$$X_{\text{sub}} \sim N_2 \left(\begin{bmatrix} 20 \\ 15 \end{bmatrix}, \begin{bmatrix} 25 & 3 \\ 3 & 9 \end{bmatrix} \right)$$

Table F.1: Theoretical and Simulated Means

Table 7: Theoretical and simulated means for the X_1, X_2 subset.

Variable	Theoretical Mean	Simulated Mean
Customer Arrival Rate (X_1)	20	19.96299879
Service Processing Time (X_2)	15	15.04168157

Table F.2: Theoretical and Simulated Covariance Matrices

Theoretical covariance matrix:

$$\Sigma_{\text{sub}} = \begin{bmatrix} 25 & 3 \\ 3 & 9 \end{bmatrix}$$

Simulated covariance matrix:

$$\hat{\Sigma}_{\text{sub}} = \begin{bmatrix} 22.27171392 & 2.53173688 \\ 2.53173688 & 9.32050792 \end{bmatrix}$$

Table F.3: Comparison of Theoretical and Simulated Covariance Values

Table 8: Comparison of covariance values for the X_1, X_2 subset.

Covariance	Theoretical	Simulated
$\text{Var}(X_1)$	25	22.27171392
$\text{Cov}(X_1, X_2)$	3	2.53173688
$\text{Var}(X_2)$	9	9.32050792

The simulated means were approximately

$$(19.963, 15.042)$$

compared with the theoretical means

$$(20, 15).$$

Summary of Relevant Numerical Values

Table 9: Summary of the main numerical results from Parts A–E and the Custom Investigation.

Part	Quantity	Value
A	μ_1	20
A	μ_2	15
A	μ_3	10
A	σ_1	5
A	σ_2	3
A	σ_3	4
A	$\text{Cov}(X_1, X_2)$	3
A	$\text{Cov}(X_1, X_3)$	15
A	$\text{Cov}(X_2, X_3)$	7.8
A	ρ_{12}	0.20
A	ρ_{13}	0.75
A	ρ_{23}	0.65
B	n	1000
B	$\bar{X}_1$	19.962999
B	$\bar{X}_2$	15.041682
B	$\bar{X}_3$	10.001995
B	s_1^2	22.271714
B	s_2^2	9.320508
B	s_3^2	14.998812
B	$\hat{\rho}_{12}$	0.176
B	$\hat{\rho}_{13}$	0.735
B	$\hat{\rho}_{23}$	0.651
C	ρ settings	0.00, 0.50, 0.90
C	$\text{Cov}(X_1, X_3)$	0, 10, 18
D	$E(Y)$	40
D	$\text{Var}(Y)$	27.8
D	$\bar{Y}$	40.044367
D	s_Y^2	27.047131
E	Degrees of freedom	3
E	90% critical value	6.2514
E	Proportion inside region	0.919
E	Percentage inside region	91.9%
F	$\boldsymbol{\mu}_{\text{sub}}$	$(20, 15)'$
F	$\bar{\mathbf{X}}_{\text{sub}}$	$(19.96299879, 15.04168157)'$